Matrix Language — Shapes, Operations, Transpose

colour key: navy = prose · teal = headings · yellow = key results · red = common traps

A matrix is a linear machine

An $m \times n$ matrix $A = (a_{ij})$ has m rows and n columns; its entry a_{ij} is in row i, column j. Multiplication by a column vector combines rows with coordinates:

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}, \quad x = \begin{pmatrix} u \\ v \end{pmatrix}, \quad Ax = \begin{pmatrix} u + 2v \\ 3u + 4v \end{pmatrix}.$$

INTUITION

Rows act; columns record

The row i of A computes the ith output as a dot product. The columns are the outputs of the coordinate basis vectors: Ae_j is column j.

Shape check before multiplying

$A_{m \times n} B_{n \times p}$ exists and has shape $m \times p$; its entry (i, j) is $\sum_{k=1}^{n} a_{ik} b_{kj}$. Inner dimensions must agree. Addition needs identical shapes. Matrix multiplication is associative and distributive, generally **not commutative**.

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} 5 & 6 \\ 7 & 8 \end{pmatrix} = \begin{pmatrix} 19 & 22 \\ 43 & 50 \end{pmatrix} \neq \begin{pmatrix} 23 & 34 \\ 31 & 46 \end{pmatrix}.$$

COMMON TRAP

Order and zero traps

AB defined does not imply BA is defined. $AB = 0$ does not force $A = 0$ or $B = 0$; singular matrices can annihilate nonzero vectors. Never cancel a matrix factor unless invertibility is established.

Identity, powers and transpose

$I_n A = A = A I_m$ when dimensions fit. $A^0 = I$ for square A; powers require a square matrix.
$(AB)^T = B^T A^T$ (reverse order),
$(A + B)^T = A^T + B^T$.

WORKED EXAMPLE

Quick transpose check

If $A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}$, then $A^T = \begin{pmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{pmatrix}$, now 3×2. Transposing swaps row/column labels, not the entries' signs.

Named structures

Square: $n \times n$. Diagonal: off-diagonal entries zero. Scalar: λI. Symmetric: $A^T = A$; skew-symmetric: $A^T = -A$, hence diagonal entries are zero over real numbers. For any real square A:

$$A = \frac{A + A^T}{2} + \frac{A - A^T}{2}$$

is the unique symmetric + skew-symmetric decomposition.

KEY RESULT

Transpose identities

$(A^T)^T = A$; $(AB)^T = B^T A^T$; $(A^{-1})^T = (A^T)^{-1}$ when inverse exists. The last follows by transposing $AA^{-1} = I$.

The determinant measures signed area / volume

For $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, $\det A = ad - bc$. The columns span a parallelogram; $|\det A|$ is its area scale, while the sign records orientation. In dimension n, determinant is the signed volume scale.

For a 3×3 matrix expand along a row/column: delete row i, column j to get minor M_{ij}; cofactor $C_{ij} = (-1)^{i+j} M_{ij}$. Then $\det A = \sum_j a_{ij} C_{ij}$. The checkerboard signs begin $+, -, +$.

$$\det \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 10 \end{pmatrix} = 1(50 - 48) - 2(40 - 42)$$
$$+3(32 - 35) = -3.$$

SHORT DERIVATION

Why the cofactor signs alternate

In the permutation formula, exchanging two columns reverses orientation. Deleting row i and column j leaves the same smaller determinant; the parity of the row/column moves contributes $(-1)^{i+j}$.

Row operations: distinguish the three

1. Swap two rows: determinant changes sign.
2. Multiply a row by c: determinant multiplies by c.
3. Add c times one row to another: determinant unchanged.

Same rules hold for columns. Thus if rows are equal/proportional, determinant is zero; a common row factor can be pulled out. Triangular determinant is product of diagonal entries.

COMMON TRAP

Scaling the whole matrix

For an $n \times n$ matrix, $\det(cA) = c^n \det A$ because each of the n rows scales. It is not merely $c\det A$ unless $n = 1$.

Structural identities

$$\det(A^T) = \det A, \qquad \det(AB) = \det A \det B$$

$$\det(A^{-1}) = \frac{1}{\det A}.$$

The inverse identity assumes $\det A \neq 0$. A is singular iff $\det A = 0$; geometrically its columns collapse into a lower-dimensional space.

WORKED EXAMPLE

Parameter test

$A = \begin{pmatrix} k & 1 \\ 2 & k \end{pmatrix}$ is invertible iff $k^2 - 2 \neq 0$, i.e. $k \neq \pm\sqrt{2}$. At either excluded value the columns are dependent and the homogeneous system has a nonzero solution.

KEY RESULT

Fast evaluation

Before expanding, look for a sparse row/column; repeated/proportional rows; row additions that create zeros; triangular form. Keep track of swaps and row scalings explicitly.

Cofactors assemble the adjoint

For square A, $C_{ij} = (-1)^{i+j} M_{ij}$ and $\operatorname{adj} A = (C_{ij})^T$: **transpose the cofactor matrix**. Cofactor expansion gives

$$A \operatorname{adj} A = \operatorname{adj} A A = (\det A) I.$$

Off-diagonal entries cancel in pairs; diagonal entries are cofactor expansions of $\det A$.

KEY RESULT

Inverse criterion and formula

A^{-1} exists exactly when $\det A \neq 0$. Then $A^{-1} = \dfrac{\operatorname{adj} A}{\det A}$. If determinant vanishes, no inverse: the adjoint formula is not a workaround.

For $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, $\operatorname{adj} A = \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$ and

$$A^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}, \quad ad - bc \neq 0.$$

WORKED EXAMPLE

Compute and verify

For $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$, determinant -2, adjoint $\begin{pmatrix} 4 & -2 \\ -3 & 1 \end{pmatrix}$, hence $A^{-1} = \begin{pmatrix} -2 & 1 \\ 3/2 & -1/2 \end{pmatrix}$. Check by multiplication: $AA^{-1} = I$.

Useful inverse laws (only when defined)

$(AB)^{-1} = B^{-1}A^{-1}$, $(A^T)^{-1} = (A^{-1})^T$, $(A^{-1})^{-1} = A$. Also $\det(A^{-1}) = 1/\det A$. To solve $AX = B$, multiply **on the left**: $X = A^{-1}B$; preserve order.

COMMON TRAP

Adjoint versus cofactor matrix

The adjoint is the transpose of the cofactor matrix. A symmetric matrix can hide this distinction because its cofactor matrix may also be symmetric; verify the definition instead of relying on that coincidence.

A practical 3×3 calculation

For $A=\begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{pmatrix}$, cofactors give $\det A = 4$ and $\operatorname{adj} A=\begin{pmatrix} 3 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 3 \end{pmatrix}$. Therefore $A^{-1}=\frac{1}{4}\operatorname{adj} A$; the quick check is $A\operatorname{adj} A = 4I$.

Encode equations as $AX = B$

For n linear equations in n unknowns, coefficient matrix A, unknown vector X, constants B. If $\det A \neq 0$, there is exactly one solution, $X = A^{-1}B$.

Cramer's rule

Replace column i of A by B to form A_i. Then $x_i = \det A_i / \det A$. This requires square A and nonzero determinant.

WORKED EXAMPLE

Two equations

$2x + y = 5$, $x - y = 1$. Here $A=\begin{pmatrix} 2 & 1 \\ 1 & -1 \end{pmatrix}$ and $\det A = -3$. Replacing columns gives determinants -6, -3, hence $(x, y) = (2, 1)$. Substitute back to check both equations.

WORKED EXAMPLE

Three unknowns

For $x + y + z = 6$, $2x - y + z = 3$, $x + 2y - z = 2$, coefficient determinant is 7. Replacing columns by $(6, 3, 2)^T$ gives determinants $7, 14, 21$, so $(x, y, z) = (1, 2, 3)$. Direct substitution checks all three equations.

COMMON TRAP

Cramer's rule does not settle singular cases

If $\det A = 0$, do not divide. The system may have no solution or infinitely many; compare the equations/ranks or row-reduce the augmented matrix.

Homogeneous systems

$AX = 0$ always has $X = 0$. A nonzero solution exists iff A is singular (for square A), since $AX = 0$ has a nontrivial null vector exactly when columns are dependent.

WORKED EXAMPLE

Parameter condition

$\begin{pmatrix} \lambda & 1 \\ 1 & \lambda \end{pmatrix}X = 0$ has nonzero solutions iff $\lambda^2 - 1 = 0$, so $\lambda = \pm 1$. For other real λ, only $X = 0$.

General consistency and rank

For m equations, n unknowns, compare coefficient rank $r(A)$ and augmented rank $r([A|B])$:

- unequal ranks: no solution;
- equal rank $r = n$: unique solution;
- equal rank $r < n$: infinitely many solutions (over $\mathbb{R}$).

The last case has $n - r$ free parameters. Rank is the number of pivots after row reduction, equivalently the largest order of a nonzero minor.

SHORT DERIVATION

Why ranks decide consistency

Row operations preserve the equation solution set. An echelon system has one pivot equation per pivot; a row $[0 \cdots 0 \mid c]$, $c \neq 0$, is a contradiction. Without contradiction, each non-pivot variable is free.

KEY RESULT

Syllabus boundary

JEE matrix methods: determinants up to order 3, adjoint/inverse, and Cramer's rule for 2- or 3-variable systems. Rank language is a useful explanatory lens; do not let it replace the required determinant conditions.

Symmetric and skew-symmetric

Real square A is symmetric when $A^T = A$; skew-symmetric when $A^T = -A$. In the latter, $a_{ii} = -a_{ii}$, so every diagonal entry is zero. Every real square matrix splits uniquely:

$$A = S + K, \quad S = \frac{A + A^T}{2}, \quad K = \frac{A - A^T}{2}.$$

WORKED EXAMPLE

Split a 2×2 matrix

For $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$, $S = \begin{pmatrix} 1 & 5/2 \\ 5/2 & 4 \end{pmatrix}$ and $K = \begin{pmatrix} 0 & -1/2 \\ 1/2 & 0 \end{pmatrix}$. Check $S^T = S$, $K^T = -K$, and $S + K = A$.

Orthogonal matrices preserve length

Real square Q is orthogonal if $Q^TQ = QQ^T = I$. Thus $Q^{-1} = Q^T$ and columns (also rows) are orthonormal. For every vector x, $\|Qx\|^2 = x^TQ^TQx = \|x\|^2$: rotations/reflections preserve lengths and angles. Taking determinants: $(\det Q)^2 = 1$, so $\det Q = \pm 1$.

$$Q = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$$

is a rotation; check orthogonality using $\cos^2\theta + \sin^2\theta = 1$.

COMMON TRAP

Transpose is not inverse in general

$A^T = A^{-1}$ only for orthogonal A. Symmetry $A^T = A$ alone does not imply orthogonality or invertibility.

Three algebraic families

Idempotent: $P^2 = P$ (projection; eigenvalue intuition only in extension). Involutory: $R^2 = I$, hence $R^{-1} = R$. Nilpotent: $N^k = 0$ for some positive integer k; nilpotent matrices are singular when positive-dimensional.

WORKED EXAMPLE

A projection

$P = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}$ has $P^2 = P$. Acting on $(x, y)^T$, it maps to $(x + y, 0)^T$; applying it again changes nothing.

KEY RESULT

Test the definition

To classify a given matrix, compute the defining product: A^T, A^2, or AA^T. A visual pattern is not a proof.

Cayley–Hamilton — Reduce Every Power

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Characteristic polynomial

For square A, define $p(\lambda) = \det(\lambda I - A)$. Eigenvalues are roots because $(A - \lambda I)v = 0$ has a nonzero vector exactly when determinant vanishes. For 2×2:

$$p(\lambda) = \lambda^2 - (\operatorname{tr} A)\lambda + \det A.$$

KEY RESULT

Cayley–Hamilton theorem

Every square matrix satisfies its characteristic polynomial: $p(A) = 0$. For order 2, $A^2 - (\operatorname{tr} A)A + (\det A)I = 0$. Therefore every higher power reduces to a linear combination of I and A.

WORKED EXAMPLE

Power reduction

For $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$, trace 5, determinant -2, so $A^2 - 5A - 2I = 0$. Then $A^2 = 5A + 2I$ and $A^3 = 27A + 10I$.

Eigenvalue invariants and similarity

$B = P^{-1}AP$ is a change of basis (similarity), when P invertible. Then $\det(\lambda I - B) = \det(\lambda I - A)$; hence trace and determinant, and the eigenvalue multiset, are unchanged. Distinct eigenvalues imply diagonalizability over a field containing all eigenvalues (for real matrices, at least over $\mathbb{C}$); repeated eigenvalues alone do not.

COMMON TRAP

Avoid an invalid substitution

Cayley–Hamilton lets us reduce powers of one fixed square matrix. It does not permit replacing a scalar variable by a matrix in an unrelated identity unless products commute and the polynomial identity applies.

Hadamard's volume bound

For columns $c_1, \ldots, c_n$ of A, $|\det A| \le \prod_j \|c_j\|$. Volume cannot exceed the rectangular box with those edge lengths; equality holds exactly when nonzero columns are pairwise orthogonal (zero-column case gives both sides zero).

SHORT DERIVATION

Two-dimensional identity

For columns $(a, c)^T, (b, d)^T$, $(ad - bc)^2 = (a^2 + c^2)(b^2 + d^2) - (ab + cd)^2$. Thus $|ad - bc|$ is at most the product of their lengths; equality iff dot product is zero.

Rank-one determinant lemma

For invertible A and compatible column vectors u, v:

$$\det(A + uv^T) = \det A(1 + v^T A^{-1} u).$$

A high-dimensional determinant update reduces to one scalar. Equivalent core identity: $\det(I + XY) = \det(I + YX)$ for compatible rectangular X, Y.

WORKED EXAMPLE

Check one update

$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$, $u = (1, 2)^T$, $v = (3, 4)^T$.

Directly $\det(A + uv^T) = -6$; the lemma gives $(-2)(1 + 2) = -6$.

Vandermonde determinant

$$\det \begin{pmatrix} 1 & x_1 & x_1^2 \\ 1 & x_2 & x_2^2 \\ 1 & x_3 & x_3^2 \end{pmatrix} = (x_2 - x_1)(x_3 - x_1)(x_3 - x_2).$$

It vanishes iff two nodes coincide, exactly when quadratic interpolation loses uniqueness. General form: $\prod_{i<j}(x_j - x_i)$.

COMMON TRAP

State the needed hypotheses

The matrix determinant lemma uses A^{-1}, so A must be invertible. The Vandermonde product depends on the stated column order $1, x, x^2$ and row order x_1, x_2, x_3; reordering nodes changes sign accordingly.

Trace: cyclic, not commutative

For square matrices A, B of the same order, $\operatorname{tr}(AB) = \operatorname{tr}(BA)$ since both equal $\sum_{i,j} a_{ij} b_{ji}$. This does **not** say $AB = BA$. More generally $\operatorname{tr}(ABC) = \operatorname{tr}(BCA)$ when dimensions permit cyclic products.

Rank–nullity

For linear map $A : \mathbb{R}^n \to \mathbb{R}^m$, $\operatorname{rank} A + \operatorname{nullity} A = n$. Row reduction exposes pivot variables and free variables; the null space is the solution space of $Ax = 0$.

WORKED EXAMPLE

Dimension count

If a 3×4 matrix has rank 2, then nullity is $4 - 2 = 2$; the homogeneous system has a two-dimensional family of solutions.

Block determinants

For square diagonal blocks A, D:

$$\det \begin{pmatrix} A & B \\ 0 & D \end{pmatrix} = \det A \det D.$$

If A is invertible (and block sizes compatible), Schur complement gives

$$\det \begin{pmatrix} A & B \\ C & D \end{pmatrix} = \det A \det(D - CA^{-1}B).$$

INTUITION

Eliminate the lower-left block

Multiply on the left by the block triangular matrix $\begin{pmatrix} I & 0 \\ -CA^{-1} & I \end{pmatrix}$, whose determinant is 1. The product has lower-left block zero and lower-right block $D - CA^{-1}B$.

WORKED EXAMPLE

Schur complement in one calculation

Let $A = I_2$, $B = (1, 2)^T$, $C = (3, 4)$, $D = (10)$. Then the Schur complement is $10 - (3, 4)(1, 2)^T = 10 - 11 = -1$, so the block determinant is -1. Directly,

$$\det \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 2 \\ 3 & 4 & 10 \end{pmatrix} = -1.$$

Boundary reminder

Eigenvalues, Cayley–Hamilton, rank–nullity, Hadamard, Vandermonde and Schur complements are useful Olympiad extensions; they are not being represented as explicit JEE syllabus requirements. Matrix algebra can still arise inside a JEE question when supplied or derivable.